

SUMBAWANGA, Tanzania, United Republic of

WMO: 638810

Lat: 7.967S

Lon: 31.630E

Elev: 6309

StdP: 11.64

Time Zone: 3.00 (E03)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	47.9	49.6	31.0	31.9	73.8	34.4	36.8	71.0	22.6	57.7	18.4	64.6	5.0	130	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	18.3	81.0	58.3	80.3	58.4	78.9	58.8	65.7	73.4	65.0	72.4	64.4	71.6	8.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	111.0	67.7	62.8	108.5	67.2	62.3	106.5	66.8	34.5	74.0	33.8	73.1	33.2	72.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	14.4	12.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			42.0	71.5	1.9	3.8	40.6	74.3	39.5	76.5	38.4	78.7	37.0	81.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.3	66.4	66.8	66.9	66.6	65.3	63.0	62.4	64.8	67.6	69.9	69.3	67.0
	DBStd	2.85	1.69	1.76	1.86	1.44	1.67	1.74	2.10	1.80	1.95	1.91	2.58	1.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	216	6	4	5	3	17	63	83	26	4	0	2	3
	CDD50	5959	508	471	525	497	473	390	385	458	528	617	579	528
	CDD65	701	49	55	65	50	25	3	3	19	82	152	131	67
	CDH74	3132	67	109	116	58	91	148	198	330	607	780	523	105
	CDH80	108	1	1	0	0	1	0	3	12	25	42	23	0
Wind	WSAvg	5.0	2.7	3.4	3.6	5.1	5.8	5.8	6.2	5.9	6.2	6.7	5.3	3.3
Precipitation	PrecAvg	43.1	8.6	7.4	7.8	3.7	0.5	0.0	0.0	0.0	0.2	1.2	4.8	8.9
	PrecMax	54.1	12.2	10.0	11.0	6.0	3.4	0.2	0.2	0.2	0.6	3.0	8.6	14.2
	PrecMin	34.2	6.3	4.5	4.7	1.3	0.0	0.0	0.0	0.0	0.0	0.3	1.8	3.2
	PrecStd	4.6	1.6	1.4	1.4	1.3	0.6	0.0	0.0	0.0	0.2	0.8	2.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.3	78.6	78.7	77.3	78.0	79.0	80.2	81.3	82.4	82.7	82.1	78.4
		MCWB	62.8	62.7	64.6	62.9	59.2	57.9	56.4	56.4	56.8	58.7	57.9	62.2
	2%	DB	75.5	76.9	76.9	75.5	76.2	77.2	77.4	79.5	80.7	81.0	80.4	76.7
		MCWB	63.0	62.9	63.4	62.6	59.7	57.4	56.0	56.3	57.7	59.2	59.1	61.8
	5%	DB	74.4	75.3	75.4	74.7	75.1	75.9	76.6	77.4	79.2	80.1	78.8	75.2
		MCWB	62.8	63.0	63.2	62.4	60.1	57.3	55.9	56.4	57.9	59.2	59.3	62.1
	10%	DB	73.0	73.7	73.9	73.4	73.6	74.3	74.7	75.7	77.8	78.6	77.2	73.5
		MCWB	62.5	62.9	63.0	62.1	59.9	57.3	55.6	56.6	57.6	59.2	59.8	62.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	66.5	66.8	65.7	64.3	61.9	60.7	61.7	63.1	65.4	65.5	66.0
		MCDB	72.4	74.1	74.5	73.5	72.9	74.3	75.5	75.5	76.2	75.7	72.9	73.3
	2%	WB	64.8	65.2	65.5	64.8	62.9	60.3	58.8	59.8	61.4	63.7	63.8	64.6
		MCDB	71.2	72.9	73.4	72.2	72.2	73.8	74.2	74.8	76.1	74.1	71.3	71.6
	5%	WB	64.0	64.5	64.6	64.0	61.9	59.2	57.7	58.5	60.0	62.6	63.0	63.7
		MCDB	70.3	71.7	72.0	71.2	71.3	73.0	73.0	73.9	75.3	74.2	70.6	70.5
	10%	WB	63.4	63.7	63.8	63.3	61.0	58.2	56.6	57.4	58.7	61.5	62.4	63.0
		MCDB	69.6	70.6	71.0	70.4	70.3	71.9	71.8	72.8	74.4	73.1	70.0	69.6
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.9	14.7	15.0	18.7	24.5	25.7	23.9	21.6	18.3	15.7	13.6
		MCDBR	16.3	17.3	16.8	16.9	20.7	26.4	28.1	26.6	23.5	20.4	17.8	15.7
		MCWBR	6.3	6.9	6.9	6.7	8.1	10.6	11.1	10.1	9.3	7.8	6.3	5.4
	5% WB	MCDBR	14.4	15.4	15.5	15.0	17.9	24.5	25.6	23.9	21.1	18.0	15.1	13.9
		MCWBR	6.3	6.9	7.2	6.8	7.7	10.9	11.2	10.1	9.2	7.8	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.354	0.343	0.331	0.303	0.296	0.291	0.304	0.340	0.371	0.387	0.351	0.338	
	taud	2.507	2.561	2.624	2.766	2.690	2.618	2.524	2.398	2.325	2.297	2.461	2.565	
	Ebn at Noon	312	314	313	314	307	304	302	297	297	297	311	317	
	Edh at Noon	36	34	32	26	27	28	31	37	42	44	37	34	
All-Sky Solar Radiation	RadAvg	1560	1715	1745	1813	1849	1828	1877	1997	2100	2085	1845	1635	
	RadStd	121	91	86	83	74	56	41	50	63	107	116	111	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page